

ON AN ESTIMATOR OF THE PROBABILITY $P(X_1 < Y, X_2 < Y, \dots, X_N < Y)$

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SUMMARY

An estimator is given for the probability that one of $(N + 1)$ mutually independent random variables, each having an absolutely continuous distribution function, is simultaneously larger (smaller) than the others. A nonparametric method of estimation is discussed, and a statistic $\hat{p}$ is shown to be consistent and the uniformly minimum variance unbiased estimator. When $N = 2$, the uniformly minimum variance unbiased estimator of the product of marginal probabilities, viz., $P(X_1 < Y) \cdot P(X_2 < Y)$ is also given.

1. Introduction

Mann and Whitney (1947) proposed a U statistic which estimates unbiasedly the probability that X is less than Y where X and Y are both random variables. The U statistic is based on the ranks of observations in the pooled sample of X's and Y's. It is also used for testing the hypothesis that X's and Y's come from the same population versus the alternative that X is stochastically smaller than Y. Van Dantzig (1951) obtained a sharp upper bound for the variance of the estimator based on the U statistic and thereby showed that Wilcoxon's two sample test studied in detail by Mann and Whitney (1947) is consistent under a considerably larger class of alternative hypotheses, namely those for which X is more likely than not to be smaller than Y.

Here, suppose that $X_1, \dots, X_N, Y$ are $(N + 1)$ mutually independent random variables (r.v.'s) with absolutely continuous distribution functions (cdf's) $F_1, \dots, F_N, G$ respectively, and

$$(1.1) \quad p = \text{Prob}(X_1 < Y, X_2 < Y, \dots, X_N < Y).$$

An example of a problem where p would be of interest is the case where a beam with strength Y is subjected to several stresses $X_1, \dots, X_N$. In this case p could logically be defined to be the reliability of the beam provided the beam fails if and only if (iff) one of

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the X 's exceeds the strength Y . Another problem of interest is to evaluate.

$$(1.2) \quad q = \text{Prob}(Y < X_1, Y < X_2, \dots, Y < X_N)$$

$$(1.3) \quad = \text{Prob}(-X_1 < -Y, -X_2 < -Y, \dots, -X_N < -Y).$$

Note that $p + q \neq 1$. Chandra (1971) and Chandra and Owen (1975) have discussed practical uses and examples for p and q and have also given Uniformly Minimum Variance (UMV) unbiased estimators of p and q in some cases where the functional form of the distributions of X 's and Y are assumed to be known.

Our principal objective is to estimate p defined above when the functional forms of the cdf's of X 's and Y are not known. However, it is assumed that the r.v.'s of X 's and Y are mutually independent and have absolutely continuous cdf's. The problem of estimating q is analogous to estimating p , in view of (1.3) and therefore is not discussed in this paper. An expression for p is given in the following lemma:

Lemma 1.1

Let $T = \max X_i$, $1 \leq i \leq N$ and $P(X_i \leq x) = F_i(x)$, $i = 1, \dots, N$. Then clearly $p = P(T < Y)$ and $H(t) = P(T \leq t) = \prod_{i=1}^N F_i(t)$. If $F_i = F$, $i = 1, \dots, N$, then $p = \int_{-\infty}^{\infty} [F(y)]^N dG(y)$ where $G(y) = P(Y \leq y)$ and if $F = G$ then $p = (N + 1)^{-1}$.

Section 2 below is concerned with estimation of p . A statistic $\hat{p}$ is proposed and is shown to be the UMV unbiased and consistent estimator of p . Also, the UMV unbiased estimator of the product of marginal probabilities namely $P(X_1 < Y) \cdot P(X_2 < Y)$ is given in Section 2. Section 3 discusses certain uses and conclusions of the results discussed in this paper.

2. An Estimator of p and Its Properties

Let $X_1, \dots, X_N, Y$ be $(N + 1)$ r.v.'s with absolutely continuous cdf's and let $X_{11}, X_{12}, \dots, X_{1n_1}$ be sample values of X_1 , $i = 1, \dots, N$ respectively and $Y_1, \dots, Y_\ell$ be sample values of Y . Let W be a count of the number of $(N + 1)$ tuples

$(X_{1i_1}, X_{2i_2}, \dots, X_{Ni_N}, Y_k)$ such that

$$X_{1i_1} < Y_k, X_{2i_2} < Y_k, \dots, X_{Ni_N} < Y_k.$$

Then $\hat{p} = W/(n_1 n_2 \dots n_N \ell)$ can be used to estimate $p = \text{Prob}(X_1 < Y, X_2 < Y, \dots, X_N < Y)$. A representation of W is as follows:

$$W = \sum_{k=1}^{\ell} U_{1k} U_{2k} \dots U_{Nk},$$

where U_{ik} = the number of $X_i < Y_k$, $i = 1, \dots, N$; $k = 1, \dots, \ell$. Thus, for example, if $N = 2$, $n_1 = 4$, $n_2 = 3$ and $\ell = 2$ and the observations are

$$X_{12} < X_{23} < Y_2 < X_{11} < X_{14} < X_{21} < Y_1 < X_{13} < X_{22},$$

then $U_{11} = 3$, $U_{12} = 1$, $U_{21} = 2$, $U_{22} = 1$ and $W = 7$ and $\hat{p} = 7/24$ is an estimate of p .

Note that the statistic $\hat{p}$ is a function of the generalized U statistic (Lehmann (1963), pp. 964-965) corresponding to the kernel

$$\phi(X_{1i_1}, \dots, X_{Ni_N}, Y_k) = \begin{cases} 1 & \text{if } X_{1i_1} < Y_k, \dots, X_{Ni_N} < Y_k \\ 0 & \text{otherwise,} \end{cases}$$

and thus $\hat{p}$ reduces to the form

$$\hat{p} = \frac{1}{n_1 \dots n_N \ell} \sum_{i_1=1}^{n_1} \dots \sum_{i_N=1}^{n_N} \sum_{k=1}^{\ell} \phi(X_{1i_1}, \dots, X_{Ni_N}, Y_k).$$

Therefore, unbiasedness, consistency and asymptotic normality of $\hat{p}$ follows from U -statistic theory. Note that if the data were in blocks $(X_{1j}, \dots, X_{Nj}, Y_j)$ then our problem reduces to a problem already solved: we can use the Mann-Whitney statistic applied to $X_{\max}$ and Y .

Theorem 2.1

The statistic $\hat{p}$ is the UMV unbiased estimator of p .

Proof:

Halmos (1946) and Fraser (1957) have shown that the statistic

$$\tau = \{X_{(11)}, X_{(12)}, \dots, X_{(1n_1)}, \dots, X_{(N1)}, X_{(N2)}, \dots, X_{(Nn_N)}, Y_{(1)}, Y_{(2)}, \dots, Y_{(\ell)}\}$$

is sufficient and complete for the class $F = \{(F_1, \dots, F_N, G): F_1, \dots, F_N, G \text{ are each absolutely continuous}\}$, where $X_{(i1)} < X_{(i2)} < \dots < X_{(in_i)}$ are ordered *within* the i -th sample; $i = 1, \dots, N$ and $Y_{(1)} < Y_{(2)} < \dots < Y_{(\ell)}$ are the ordered Y 's. Since $\hat{p}$ is a function of τ and is unbiased, it follows that $\hat{p}$ is the UMV unbiased estimator of p .

Theorem 2.2

The statistic $\hat{p}$ is a consistent estimator of p .

Proof:

The proof is straightforward.

Theorem 2.3

When $N = 2$, the statistic

$$\hat{\Theta} = \frac{\sum_{k=1}^{\ell} U_{1k} \cdot \sum_{k=1}^{\ell} U_{2k} - W}{n_1 n_2^{\ell(\ell-1)}}$$

is the UMV unbiased estimator of $\Theta = p_1 p_2$, where $p_i = P(X_i < Y)$, $i = 1, 2$.

Proof:

First, note that

$$\begin{aligned} \sum_{k=1}^{\ell} U_{1k} \cdot \sum_{k=1}^{\ell} U_{2k} &= \sum_{i_1=1}^{n_1} \sum_{k=1}^{\ell} \phi(X_{1i_1}, Y_k) \cdot \sum_{i_2=1}^{n_2} \sum_{k=1}^{\ell} \phi(X_{2i_2}, Y_k) \\ &= \sum_{i_1=1}^{n_1} \sum_{i_2=1}^{n_2} \sum_{k=1}^{\ell} \phi(X_{1i_1}, X_{2i_2}, Y_k) \\ &\quad + \sum_{i_1=1}^{n_1} \sum_{i_2=1}^{n_2} \sum_{k=1}^{\ell} \sum_{\substack{k'=1 \\ k \neq k'}}^{\ell} \phi(X_{1i_1}, Y_k) \cdot \phi(X_{2i_2}, Y_{k'}) \end{aligned}$$

Note that the first term is W and by inspection the expected value of the second term is $n_1 n_2^{\ell(\ell-1)} p_1 p_2$. Hence it follows that $E(\hat{\Theta}) = \theta$. Moreover, $\hat{\Theta}$ is a function of the complete sufficient statistic τ given in Theorem 2.1, it follows that $\hat{\Theta}$ is the UMV unbiased estimator of θ .

Corollary 2.1

If $G = F_1 = F_2$, we have $p = \frac{1}{3}$, $p_1 = \frac{1}{2}$ and $p_2 = \frac{1}{2}$, so that

$$E \sum_{k=1}^{\ell} U_{1k} \cdot \sum_{k=1}^{\ell} U_{2k} = \frac{1}{12} n_1 n_2^{\ell} + \frac{1}{4} n_1 n_2^{\ell^2}$$

which is the same result as derived by D.R. Whitney (1951).

3. Uses and Conclusion

The proposed statistics can be used to estimate the reliability of a component from stress-strength relationships using the probability arguments of a certain physical model

of failure, where a component is subject to several stresses (strengths) whereas its strength (stress) is a single random variable. Such situations arise where at any moment the applied stress (or load or force) may not be measured in terms of a single random variable, but can be measured in terms of several random variables, say $X_1, \dots, X_N$, and the resistance is still a single random variable, say Y .

In many cases, a component does not fail if all the different stresses $X_1, \dots, X_N$ are simultaneously less than the resistance Y . Then, a measure of reliability of a component subject to several different stresses during a period $[0, T]$ is taken to be the probability that all the different stresses are simultaneously less than the resistance during the entire interval, i.e. $p = P(X_1 < Y, \dots, X_N < Y)$.

Church and Harris (1971) considered the example of missile flights, where the several different stresses are propulsive force, angle of elevation, changes in atmospheric condition, and so on. They assumed that all of these different stresses have known distributions, and a mathematical model is available to compute the "over-all" stress, and also the distribution of "over-all" stress is known. The statistics given here are applicable to this situation or to one similar. The difficulty, however, lies in the fact that the assumptions might not be valid in a given physical situation.

The statistic W can also be used to test the hypothesis that the random variables $X_1, \dots, X_N, Y$ have the same continuous cdf against the alternative that Y is simultaneously larger than the N random variables $X_1, \dots, X_N$, i.e., Y is larger than the maximum of $X_1, \dots, X_N$.

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